

ITVSA0-44-Formative Assessment-Mowbray Campus-5737241

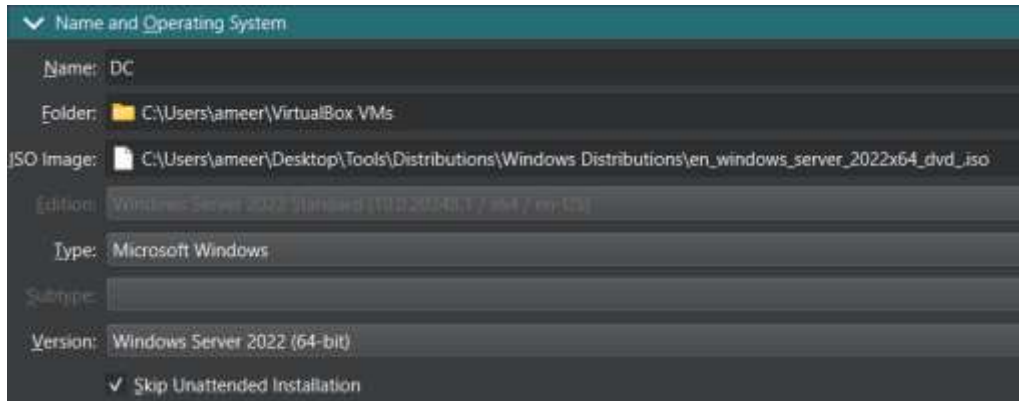
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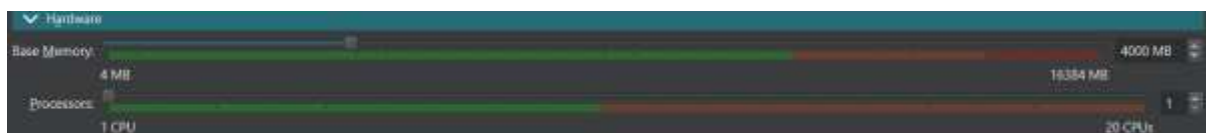
Question 1: Configurations

DC (Windows Server 2022) Configuration:

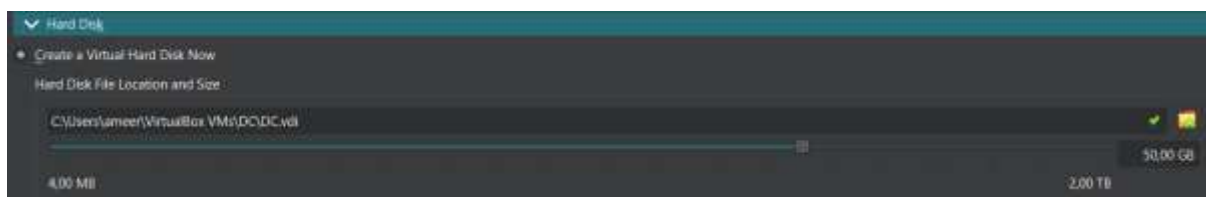
Create new VM named 'DC', with a Windows Server 2022 ISO.



Add 4GB of RAM.



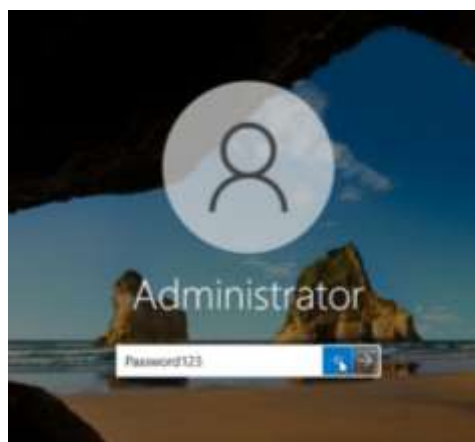
Add 50GB of storage.



1.1 Server Installation and Initial Setup

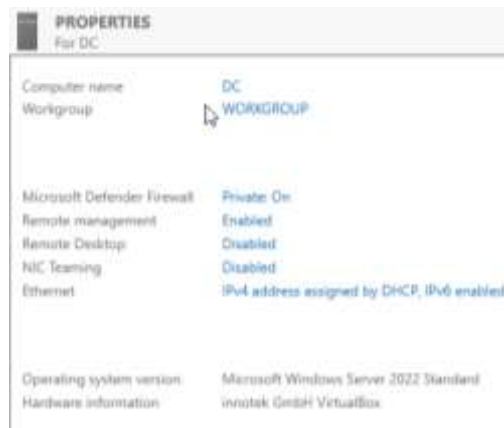
Install Windows Server 2022 Standard (Select the Desktop Experience version)

1.1a Set Password: Configure the Administrator password as **Password123**.



1.1b Change Computer Name:

Rename the server from its default name to DC.



Restart the server to apply the name change.

1.1.c TCP/IP Configuration

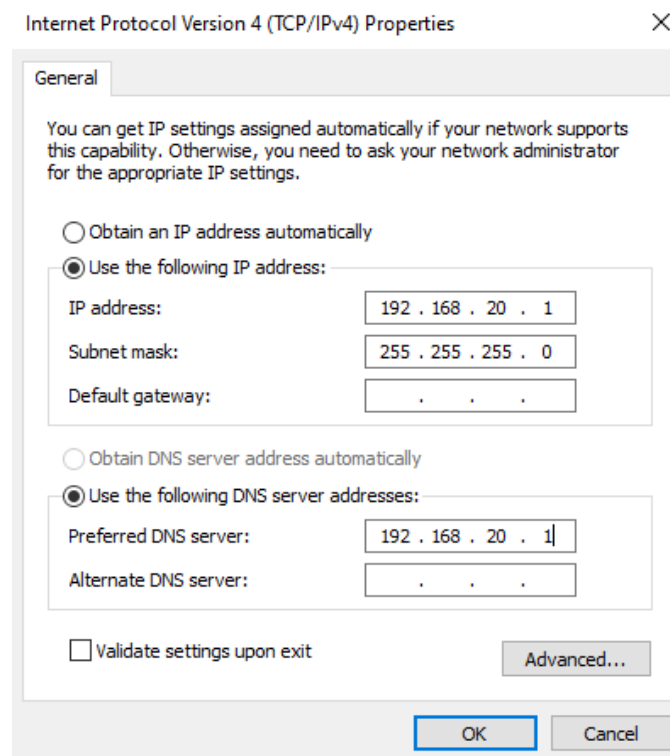
Configure Static IP: Open Network Adapter Properties (ncpa.cpl).

IP Address: 192.168.20.1

Subnet Mask: 255.255.255.0

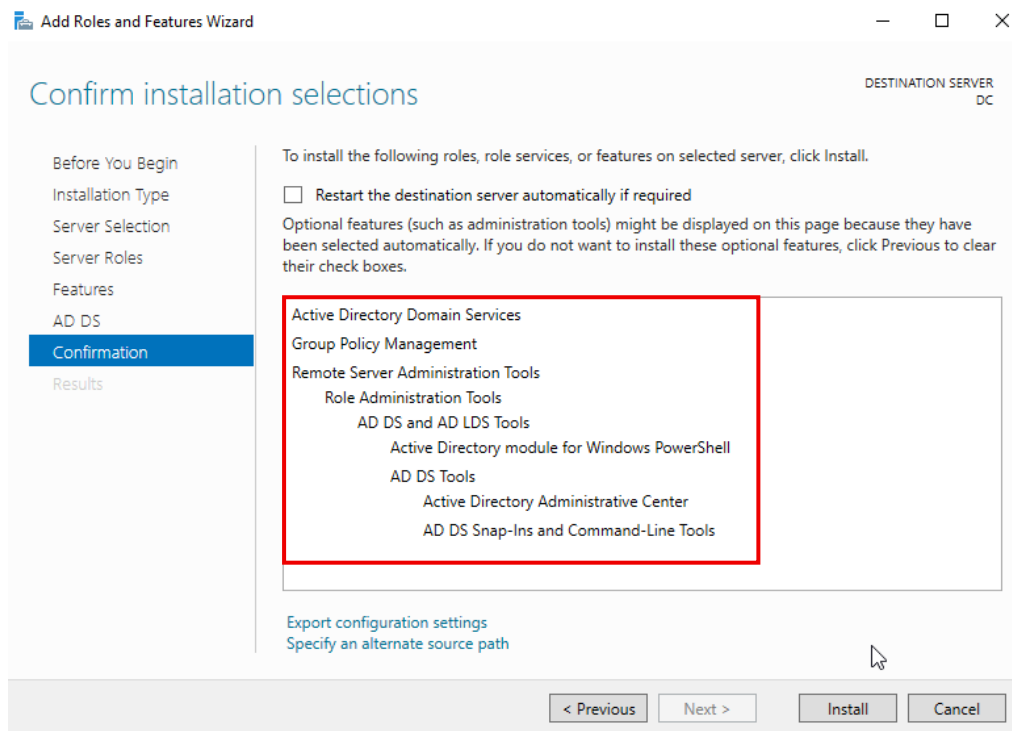
Default Gateway: (Leave Blank)

DNS Server address: 192.168.20.1 (Self-referencing for future AD/DNS role).

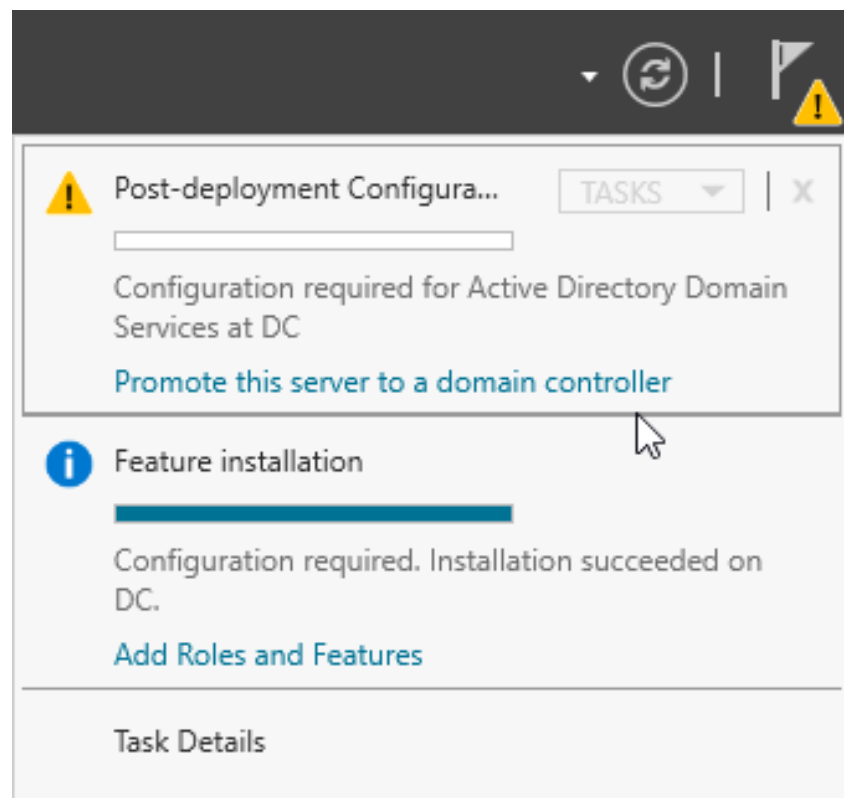


1.1.d Promote to Domain Controller

Install AD DS Role: In Server Manager, add the Active Directory Domain Services (AD DS) role.



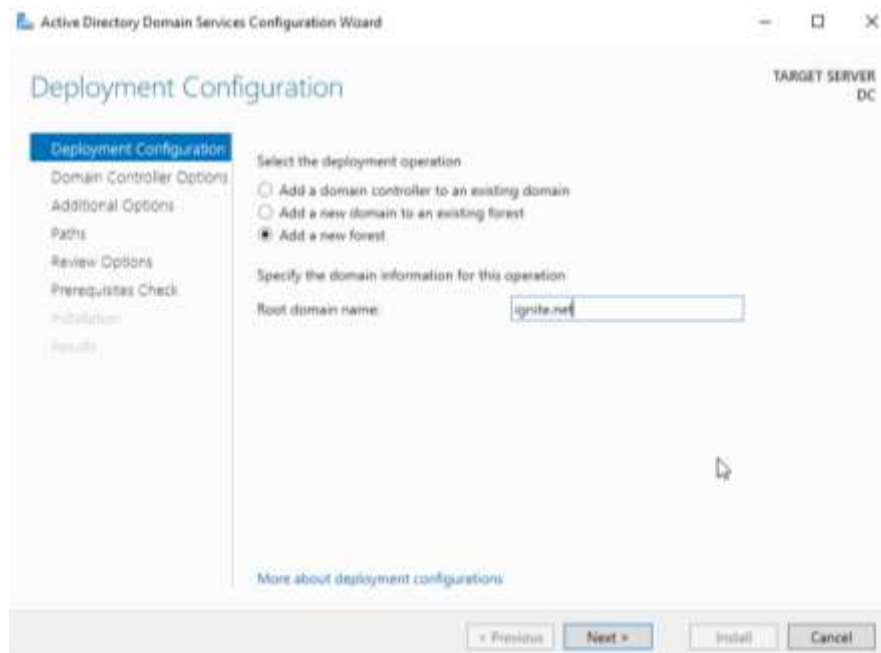
Promote Server: Click the yellow notification flag and select Promote this server to a domain controller.



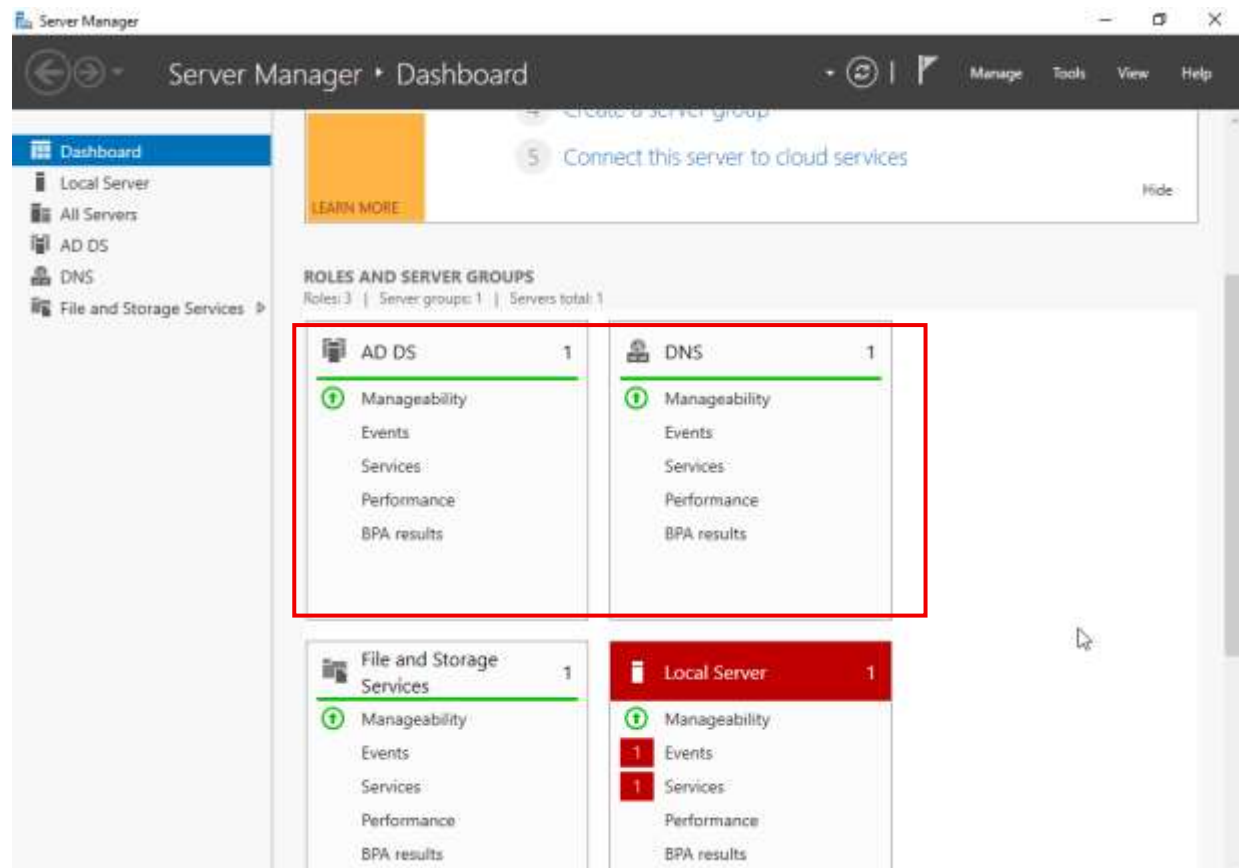
Deployment Configuration:

Select Add a new forest.

Root domain name: ignite.net



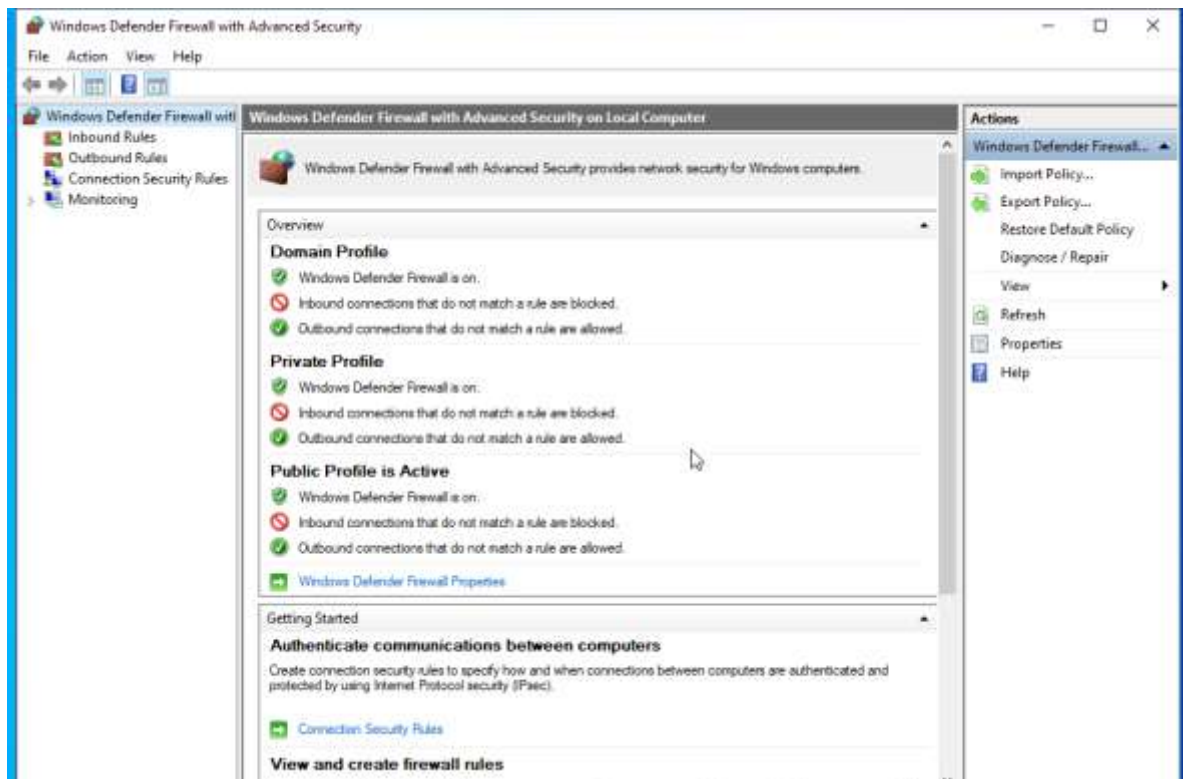
Complete the promotion process. The server will automatically restart.



1.1.e Firewall Configuration

Allow File and Printer Sharing:

Open Windows Defender Firewall with Advanced Security.



Navigate to Inbound Rules.

Enable all File and Printer Sharing rules for All Profiles (Domain, Private, Public).

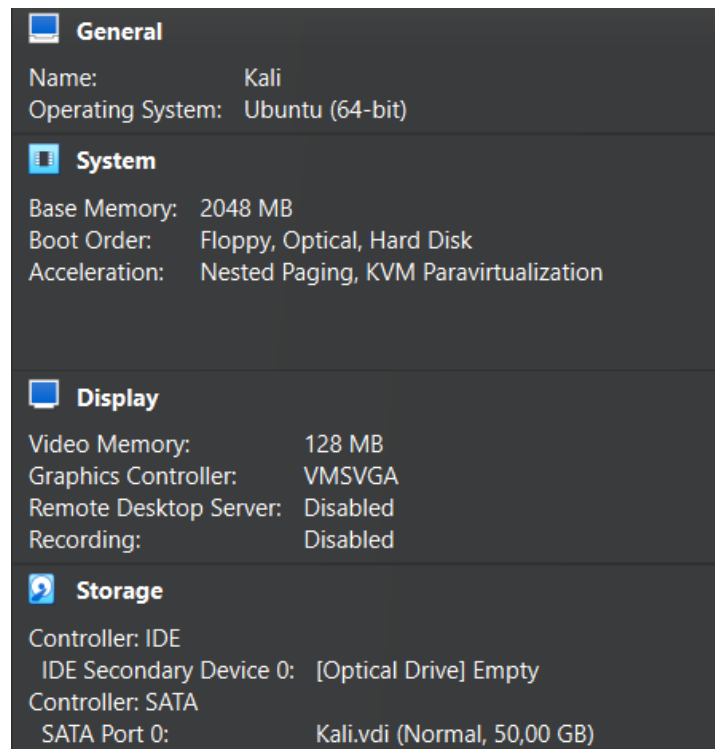
Inbound Rules					
Name	Group	Profile	Enabled	Action	
✓ DIAL protocol server (HTTP-In)	DIAL protocol server	Private	Yes	Allow	
✓ DIAL protocol server (HTTP-In)	DIAL protocol server	Domain	Yes	Allow	
Distributed Transaction Coordinator (RPC)	Distributed Transaction Co...	All	No	Allow	
Distributed Transaction Coordinator (RPC)	Distributed Transaction Co...	All	No	Allow	
Distributed Transaction Coordinator (TCP)	Distributed Transaction Co...	All	No	Allow	
✓ DNS (TCP, Incoming)	DNS Service	All	Yes	Allow	
✓ DNS (UDP, Incoming)	DNS Service	All	Yes	Allow	
✓ RPC (TCP, Incoming)	DNS Service	All	Yes	Allow	
✓ RPC Endpoint Mapper (TCP, Incoming)	DNS Service	All	Yes	Allow	
✓ File and Printer Sharing (Echo Request - I...)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (Echo Request - I...)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (LLMNR-UDP-In)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (NB-Datagram-In)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (NB-Name-In)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (NB-Session-In)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (SMB-In)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (Spooler Service - ...)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (Spooler Service - ...)	File and Printer Sharing	All	Yes	Allow	
✓ File and Printer Sharing (SMB-QUIC-In)	File and Printer Sharing over...	All	Yes	Allow	
✓ File and Printer Sharing over SMBDirect (I...	File and Printer Sharing over...	All	Yes	Allow	
✓ File Replication (RPC)	File Replication	All	Yes	Allow	
✓ File Replication (RPC-EPMAP)	File Replication	All	Yes	Allow	
✓ File Server Remote Management (DCOM-...	File Server Remote Manage...	All	Yes	Allow	
✓ File Server Remote Management (SMB-In)	File Server Remote Manage...	All	Yes	Allow	
✓ File Server Remote Management (WMI-In)	File Server Remote Manage...	All	Yes	Allow	
✓ iSCSI Service (TCP-In)	iSCSI Service	All	No	Allow	
✓ Kerberos Key Distribution Center - PCR (I...	Kerberos Key Distribution C...	All	Yes	Allow	
✓ Kerberos Key Distribution Center - PCR (I...	Kerberos Key Distribution C...	All	Yes	Allow	
✓ Kerberos Key Distribution Center (TCP-In)	Kerberos Key Distribution C...	All	Yes	Allow	

This opens a known attack vector (SMB/CIFS) necessary for lab testing.

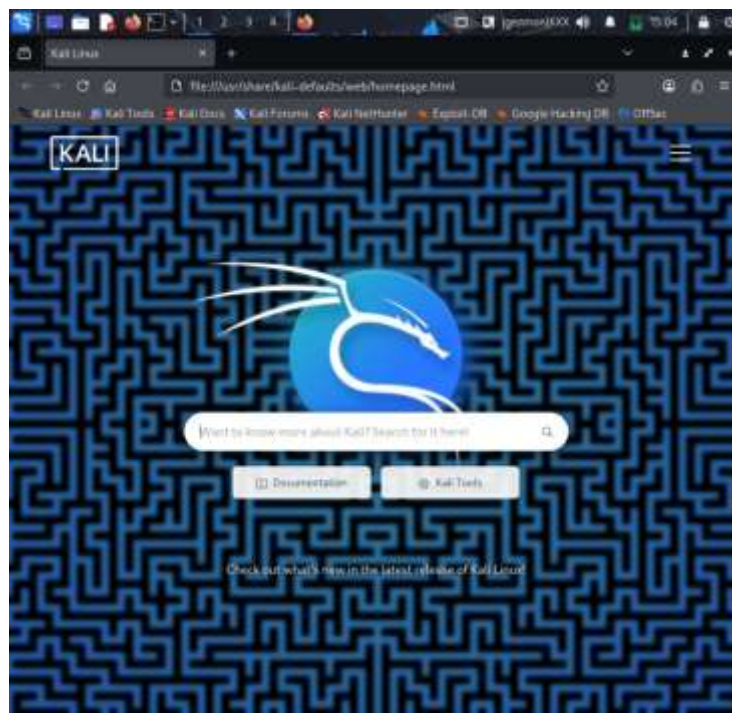
1.2 Kali Linux Configuration

Install Kali Linux:

Add 2GB RAM and 50GB storage:



1.2a Password: Password 'kali'



1.2.b TCP/IP Configuration

Configure Static IP (CLI): Edit the network configuration files.

Step 1: Identify the Network Interface

"ip a"

```
(princebardock@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 172.16.1.2/24 brd 172.16.1.255 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0f:49:c2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.9/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 370sec preferred_lft 370sec
    inet6 fe80::a00:27ff:fe0f:49c2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Step 2: Backup the Configuration File

"sudo cp /etc/network/interfaces /etc/network/interfaces.bak"

```
(princebardock@kali)-[~]
$ sudo cp /etc/network/interfaces /etc/network/interfaces.bak
[sudo] password for princebardock:
```

Step 3: Edit the Network Configuration File

"sudo nano /etc/network/interfaces"

```
(princebardock@kali)-[~]
$ sudo nano /etc/network/interfaces
```

Step 4: Input the Static Configuration

Manual (Static)

IP Address: 192.168.20.2

Subnet Mask: 255.255.255.0 (or /24)

DNS Server: 192.168.20.1 (To resolve the DC/Domain name).

```
GNU nano 8.6 /etc/network/interfaces *
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet static
    address 192.168.20.2
    netmask 255.255.255.0
    gateway 192.168.20.1
```

Verify and Test Connectivity:

Run *'ip a'* in a terminal to confirm the static IP.

```
(princebardock@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 192.168.20.2/24 brd 192.168.20.255 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0f:49:c2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.2/24 brd 192.168.20.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::dda:7ab1:7008:7a31/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Test connectivity to the DC:

ping 192.168.20.1 (IP Connectivity)

```
(princebardock@kali)-[~]
$ ping -c 4 192.168.20.1
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=64 time=0.027 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=64 time=0.043 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=64 time=0.027 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=64 time=0.050 ms

— 192.168.20.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3057ms
rtt min/avg/max/mdev = 0.027/0.036/0.050/0.010 ms
```

Question 2: Performing Reconnaissance & Analysing Data with Wireshark

Step 1: Install and Start Packet Capture on DC

2.1.1 Install Wireshark:

Download and install Wireshark on the Windows Server (DC).

Ensure the required packet capture driver (Npcap) is installed.



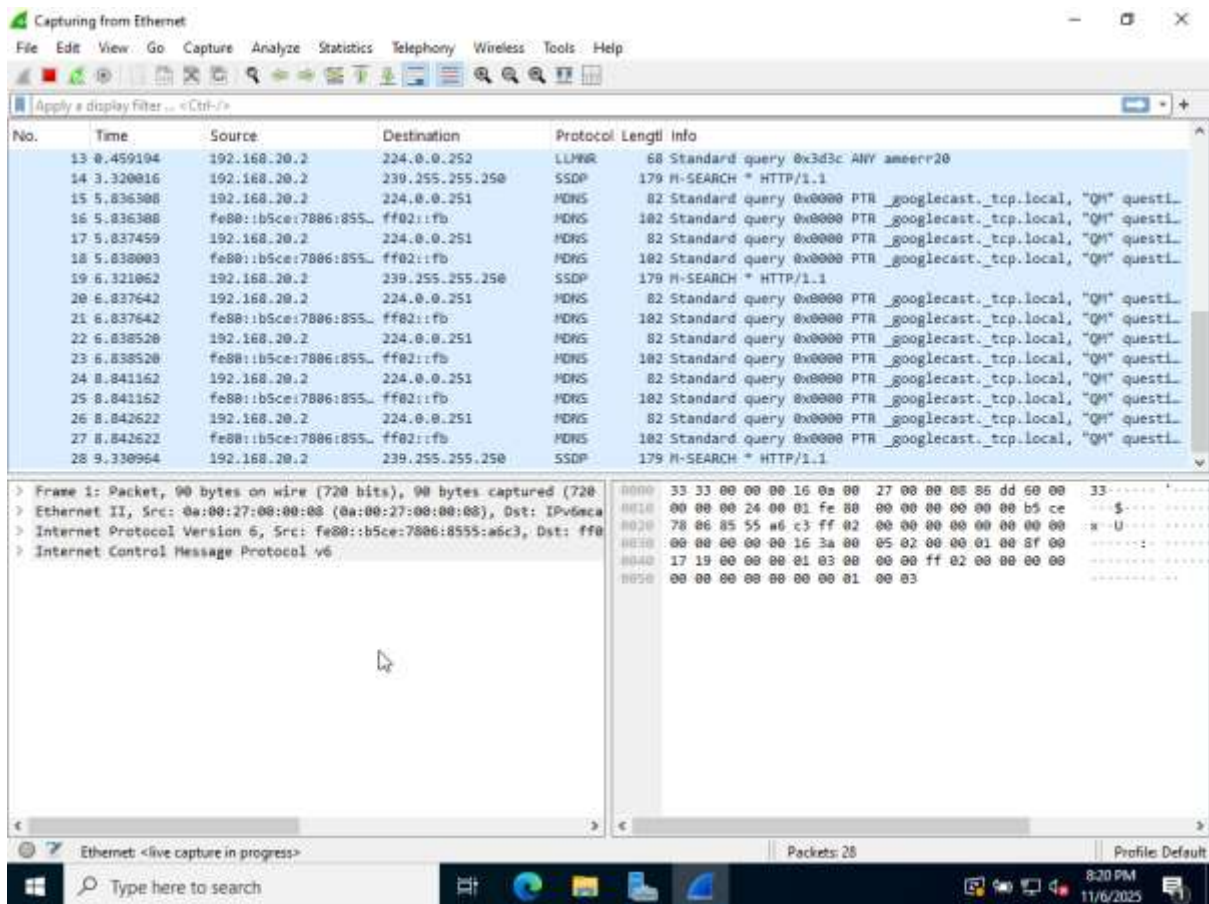
2.1.2 Start Packet Capture:

Launch Wireshark.

Select the correct network interface (NIC 1: connected to the 192.168.20.0/24 private network).

Start the live packet capture.





Step 2: Perform Reconnaissance Scans from Kali

2.2.a Run a Ping Scan (Host Discovery):

Open a terminal on Kali Linux.

Execute the Nmap command for a non-port scanning host discovery.

```
"nmap -sn 192.168.20.0/24"
```

```
(princebardock@kali)~$ nmap -sn 192.168.20.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 16:26 PST
Nmap scan report for 192.168.20.0
Host is up (0.00098s latency).
MAC Address: 0A:00:27:00:00:41 (Unknown)
Nmap scan report for 192.168.20.2
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 7.42 seconds
```

2.2.b Run a Full Scan on the Server (Service Enumeration):

Execute the Nmap command for a full TCP connect scan against the DC's IP.

```
"nmap -p- 192.168.20.1"
```



```

(princebardock@kali)-[~]
$ nmap -p- 192.168.20.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-07 16:27 PST
Nmap scan report for 192.168.20.1
Host is up (0.0017s latency).
Not shown: 65507 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsdapi
5985/tcp  open  wsman
9389/tcp  open  adws
47001/tcp open  winrm
49664/tcp open  unknown
49665/tcp open  unknown
49666/tcp open  unknown
49667/tcp open  unknown
49668/tcp open  unknown
49670/tcp open  unknown
49674/tcp open  unknown
49675/tcp open  unknown
49682/tcp open  unknown
49683/tcp open  unknown
49721/tcp open  unknown
49973/tcp open  unknown
MAC Address: 08:00:27:57:99:62 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 25.82 seconds

```

Stop Capture: Immediately stop the packet capture on the DC's Wireshark instance.

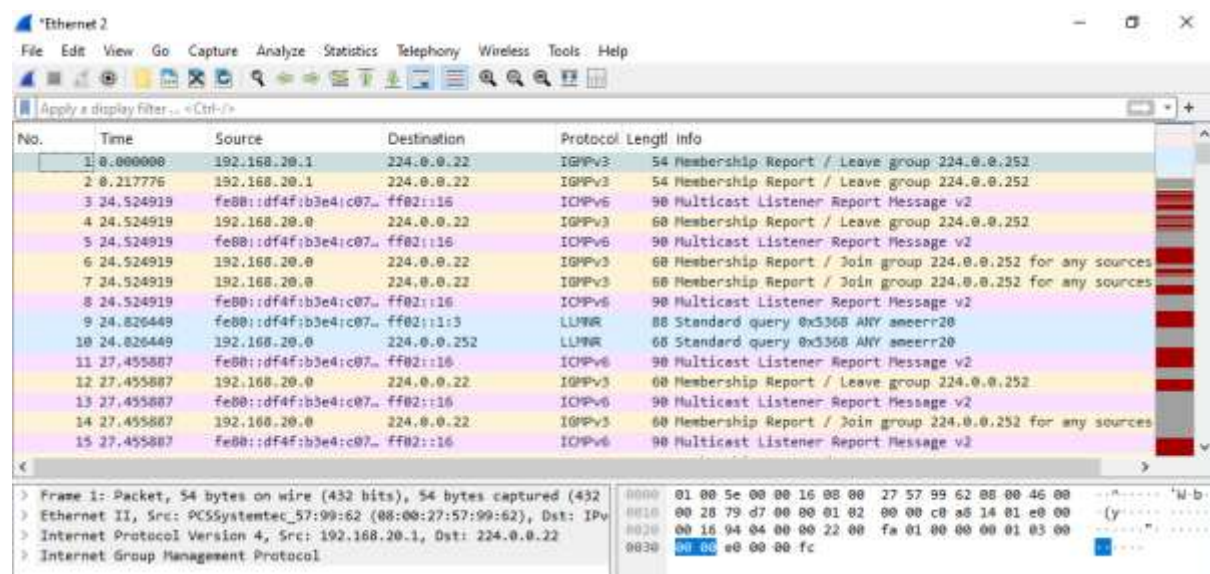
Step 3: Analyze Captured Packets (2.3) and List

2.3 Analyze the Captured Packets:

Sort the packets by time to see the chronological order of the attacks.

Observe the different protocols involved: ICMP (Ping Scan), TCP (Full Scan).

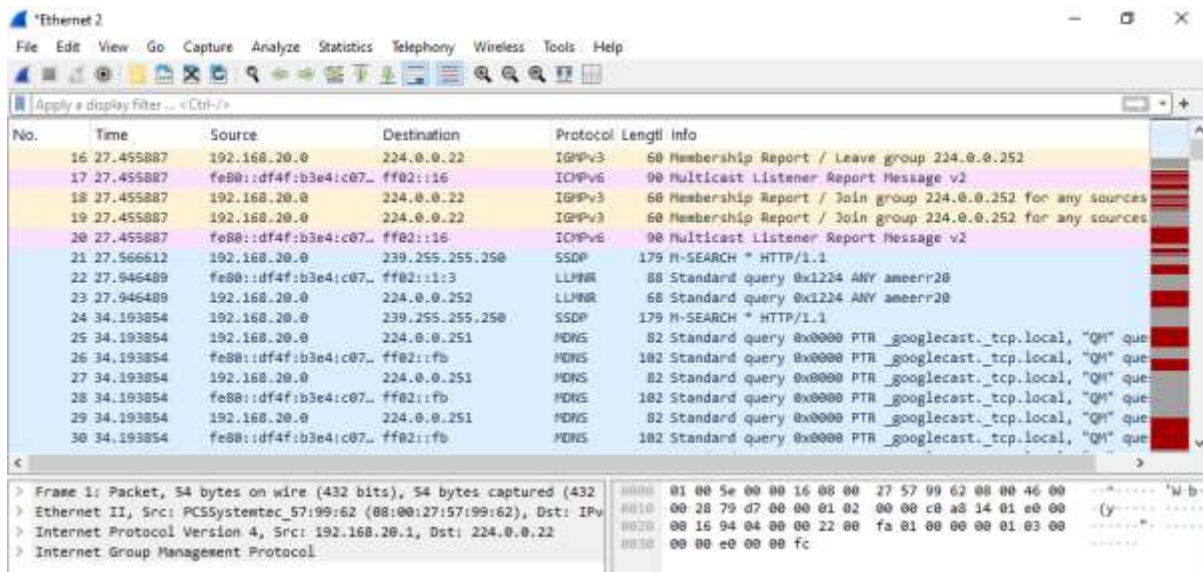
ICMP Echo Request (From Ping Scan, Kali → DC).



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.20.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
2	0.217776	192.168.20.1	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
3	24.524919	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
4	24.524919	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Leave group 224.0.0.252
5	24.524919	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
6	24.524919	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
7	24.524919	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
8	24.524919	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
9	24.826448	fe80::df4f:b3e4:c07...	ff02::1:3	LLNR	88	Standard query 0x5368 ANY ameerr20
10	24.826448	192.168.20.0	224.0.0.252	LLNR	68	Standard query 0x5368 ANY ameerr20
11	27.455887	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
12	27.455887	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Leave group 224.0.0.252
13	27.455887	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
14	27.455887	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
15	27.455887	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2

Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432) on interface Ethernet II, Src: PCSSystemtec_57:99:62 (08:00:27:57:99:62), Dst: IPv4 Internet Protocol Version 4, Src: 192.168.20.1, Dst: 224.0.0.22, Internet Group Management Protocol

ICMP Echo Reply (From Ping Scan, DC → Kali).



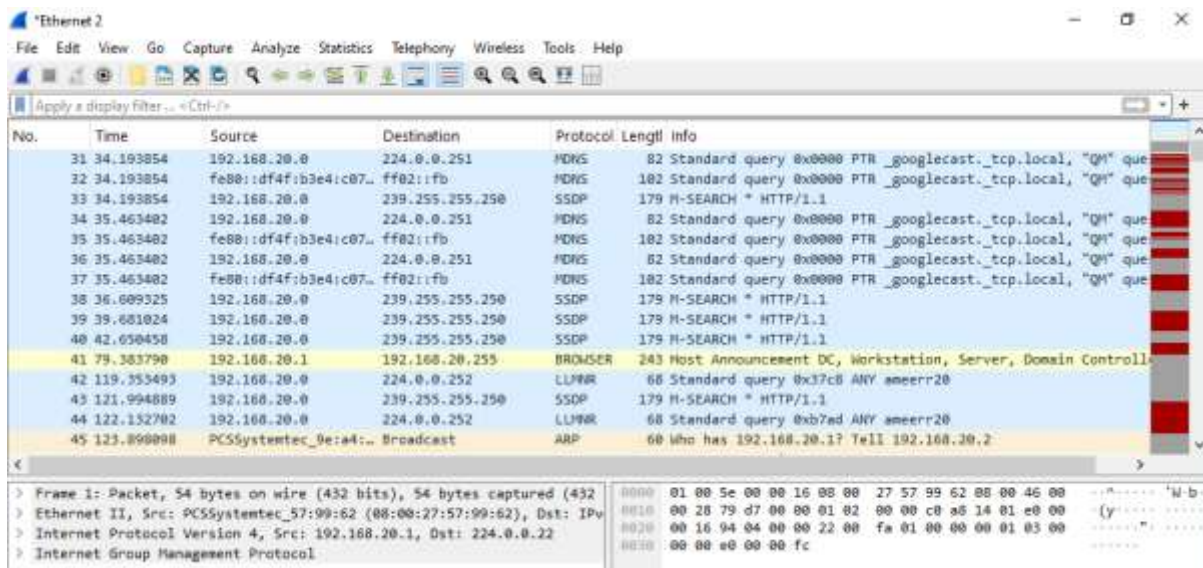
Wireshark packet capture showing network traffic. The main display shows a list of packets with columns: No., Time, Source, Destination, Protocol, Length, and Info. The packet list includes:

No.	Time	Source	Destination	Protocol	Length	Info
16	27.455887	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Leave group 224.0.0.252
17	27.455887	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
18	27.455887	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
19	27.455887	192.168.20.0	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.252 for any sources
20	27.455887	fe80::df4f:b3e4:c07...	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
21	27.566612	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
22	27.946489	fe80::df4f:b3e4:c07...	ff02::1:3	LLMNR	88	Standard query 0x1224 ANY ameerr20
23	27.946489	192.168.20.0	224.0.0.252	LLMNR	68	Standard query 0x1224 ANY ameerr20
24	34.193854	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
25	34.193854	192.168.20.0	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
26	34.193854	fe80::df4f:b3e4:c07...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
27	34.193854	192.168.20.0	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
28	34.193854	fe80::df4f:b3e4:c07...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
29	34.193854	192.168.20.0	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
30	34.193854	fe80::df4f:b3e4:c07...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que

The packet details pane shows the selected packet (No. 30) with the following information:

- Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432)
- Ethernet II, Src: PCSSystemtec_57:99:62 (08:00:27:57:99:62), Dst: IPv
- Internet Protocol Version 4, Src: 192.168.20.1, Dst: 224.0.0.22
- Internet Group Management Protocol

TCP SYN (From Full Scan, Kali → DC).



Wireshark packet capture showing network traffic. The main display shows a list of packets with columns: No., Time, Source, Destination, Protocol, Length, and Info. The packet list includes:

No.	Time	Source	Destination	Protocol	Length	Info
31	34.193854	192.168.20.0	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
32	34.193854	fe80::df4f:b3e4:c07...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
33	34.193854	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
34	35.463402	192.168.20.0	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
35	35.463402	fe80::df4f:b3e4:c07...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
36	35.463402	192.168.20.0	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
37	35.463402	fe80::df4f:b3e4:c07...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" que
38	36.609325	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
39	39.681024	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
40	42.650450	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
41	79.383790	192.168.20.1	192.168.20.255	BROWSER	243	Host Announcement DC, Workstation, Server, Domain Controll
42	119.353493	192.168.20.0	224.0.0.252	LLMNR	68	Standard query 0x37c8 ANY ameerr20
43	121.994889	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
44	122.132702	192.168.20.0	224.0.0.252	LLMNR	68	Standard query 0xb7ed ANY ameerr20
45	123.890098	PCSSystemtec_0e:a4:...	Broadcast	ARP	60	Who has 192.168.20.1? Tell 192.168.20.2

The packet details pane shows the selected packet (No. 45) with the following information:

- Frame 1: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432)
- Ethernet II, Src: PCSSystemtec_57:99:62 (08:00:27:57:99:62), Dst: IPv
- Internet Protocol Version 4, Src: 192.168.20.1, Dst: 224.0.0.22
- Internet Group Management Protocol

TCP SYN/ACK (From Full Scan, DC → Kali).

Wireshark capture showing network traffic. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
46	123.000133	PCSSystemtec_57:99:...	PCSSystemtec_9e:a4:...	ARP	42	192.168.20.1 is at 08:00:27:57:99:62
47	124.996382	192.168.20.0	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
48	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
49	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
50	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
51	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
52	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 3305 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
53	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
54	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 507 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
55	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
56	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
57	125.059049	192.168.20.2	192.168.20.1	TCP	60	41663 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
58	125.060140	192.168.20.1	192.168.20.2	TCP	58	53 → 41663 [SYN, ACK] Seq=0 Ack=1 Win=64340 Len=0 MSS=1460
59	125.060667	192.168.20.1	192.168.20.2	TCP	54	256 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
60	125.061017	192.168.20.1	192.168.20.2	TCP	54	1025 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Packet details for Frame 1 (Packet, 54 bytes on wire (432 bits), 54 bytes captured (432) on interface 0):

- Ethernet II, Src: PCSSystemtec_57:99:62 (08:00:27:57:99:62), Dst: IPv
- Internet Protocol Version 4, Src: 192.168.20.1, Dst: 224.0.0.22
- Internet Group Management Protocol

Packet bytes (hex): 0000 01 00 5e 00 00 16 08 00 27 57 99 62 08 00 46 00 0010 00 28 79 d7 00 00 01 02 00 00 c0 a5 14 01 e0 000020 00 16 94 04 00 00 22 00 fa 01 00 00 00 01 03 000030 00 00 e0 00 00 fc

TCP RST (From Full Scan, DC → Kali).

Wireshark capture showing network traffic. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
57	125.063021	192.168.20.1	192.168.20.2	TCP	54	80 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
60	125.065992	192.168.20.2	192.168.20.1	TCP	60	41663 → 8000 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
69	125.065992	192.168.20.2	192.168.20.1	TCP	60	41663 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
70	125.065979	192.168.20.1	192.168.20.2	TCP	54	8000 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
71	125.066312	192.168.20.1	192.168.20.2	TCP	54	5900 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
72	125.103998	192.168.20.2	192.168.20.1	TCP	60	41663 → 53 [RST] Seq=1 Win=0 Len=0
73	125.106490	192.168.20.2	192.168.20.1	TCP	60	41663 → 445 [RST] Seq=1 Win=0 Len=0
74	125.107292	192.168.20.2	192.168.20.1	TCP	60	41663 → 135 [RST] Seq=1 Win=0 Len=0
75	125.107070	192.168.20.2	192.168.20.1	TCP	60	41663 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
76	125.107070	192.168.20.2	192.168.20.1	TCP	60	41663 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
77	125.107070	192.168.20.2	192.168.20.1	TCP	60	41663 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
78	125.107070	192.168.20.2	192.168.20.1	TCP	60	41663 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
79	125.107090	192.168.20.1	192.168.20.2	TCP	54	113 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
80	125.108542	192.168.20.1	192.168.20.2	TCP	50	139 → 41663 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460
81	125.108541	192.168.20.1	192.168.20.2	TCP	54	21 → 41663 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

Packet details for Frame 1 (Packet, 54 bytes on wire (432 bits), 54 bytes captured (432) on interface 0):

- Ethernet II, Src: PCSSystemtec_57:99:62 (08:00:27:57:99:62), Dst: IPv
- Internet Protocol Version 4, Src: 192.168.20.1, Dst: 224.0.0.22
- Internet Group Management Protocol

Packet bytes (hex): 0000 01 00 5e 00 00 16 08 00 27 57 99 62 08 00 46 00 0010 00 28 79 d7 00 00 01 02 00 00 c0 a5 14 01 e0 000020 00 16 94 04 00 00 22 00 fa 01 00 00 00 01 03 000030 00 00 e0 00 00 fc

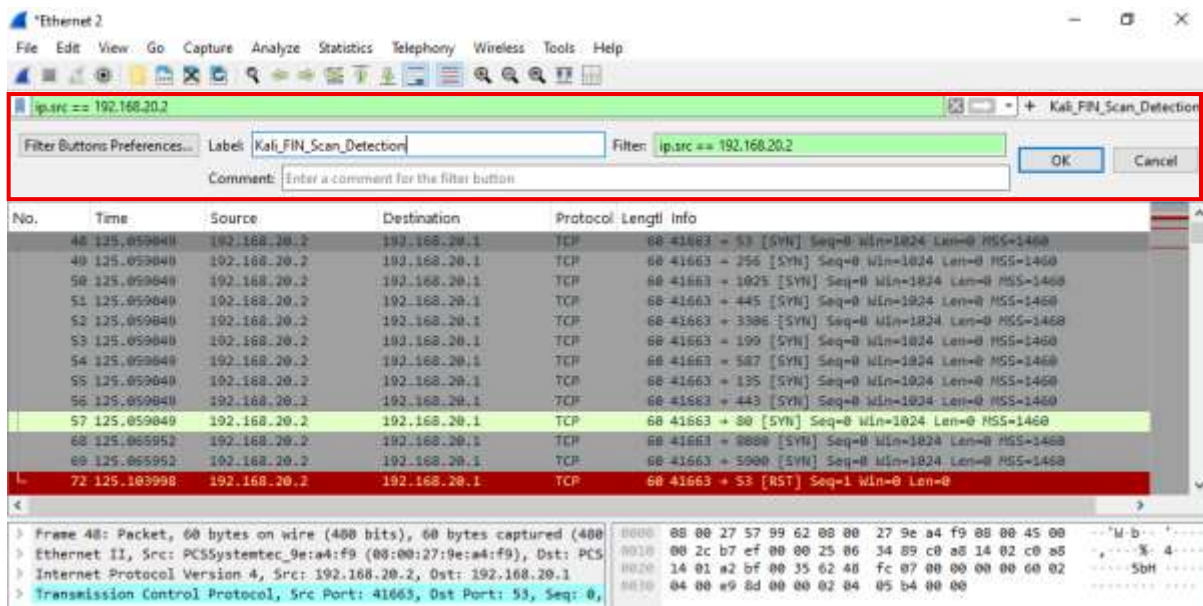
Step 4: Filter and Save Wireshark Filters

The source IP for the attacker is 192.168.20.2 (Kali).

2.5.a Filter for TCP FIN Flag

Enter the following display filter and press Enter.

"ip.src == 192.168.20.2 and tcp.flags.fin == 1"

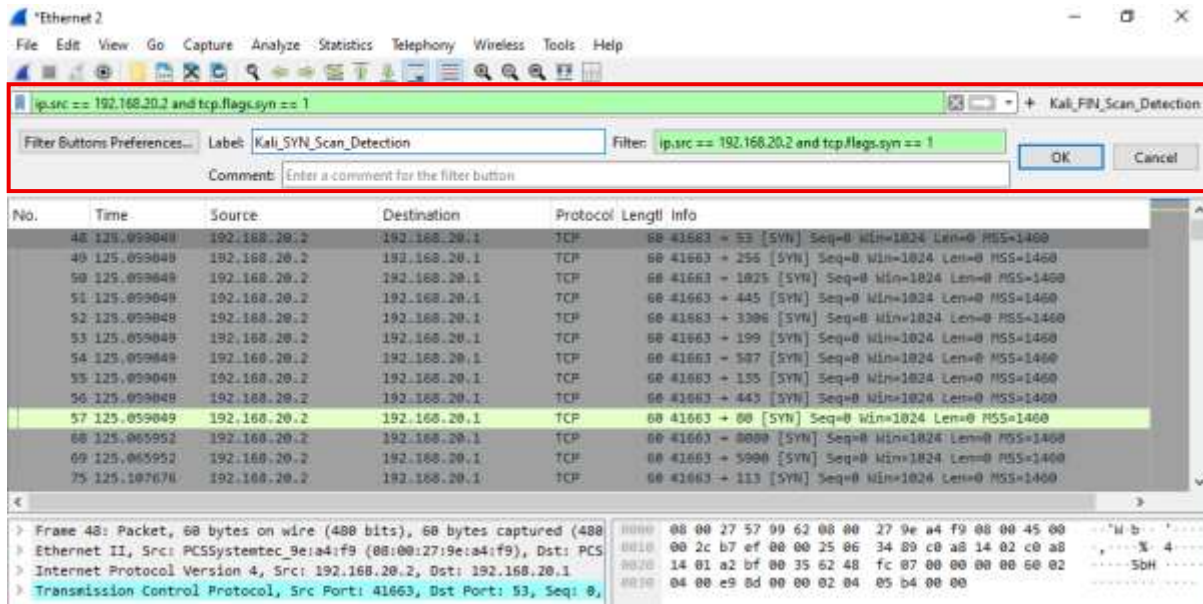


Click the + icon next to the filter bar. Save the filter with a descriptive name (Kali_FIN_Scan_Detection).

2.5.b Filter for TCP SYN Flag

Enter the following display filter and press Enter.

"ip.src == 192.168.20.2 and tcp.flags.syn == 1"



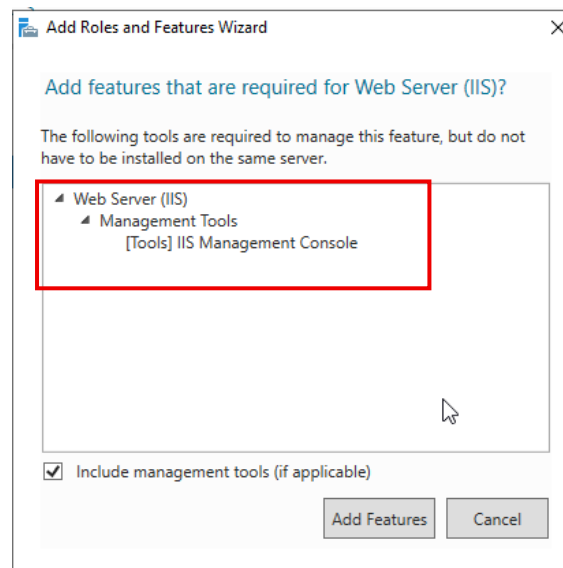
Click the + icon next to the filter bar. Save the filter with a descriptive name (Kali_SYN_Scan_Detection).

Question 3: FTP Brute Force

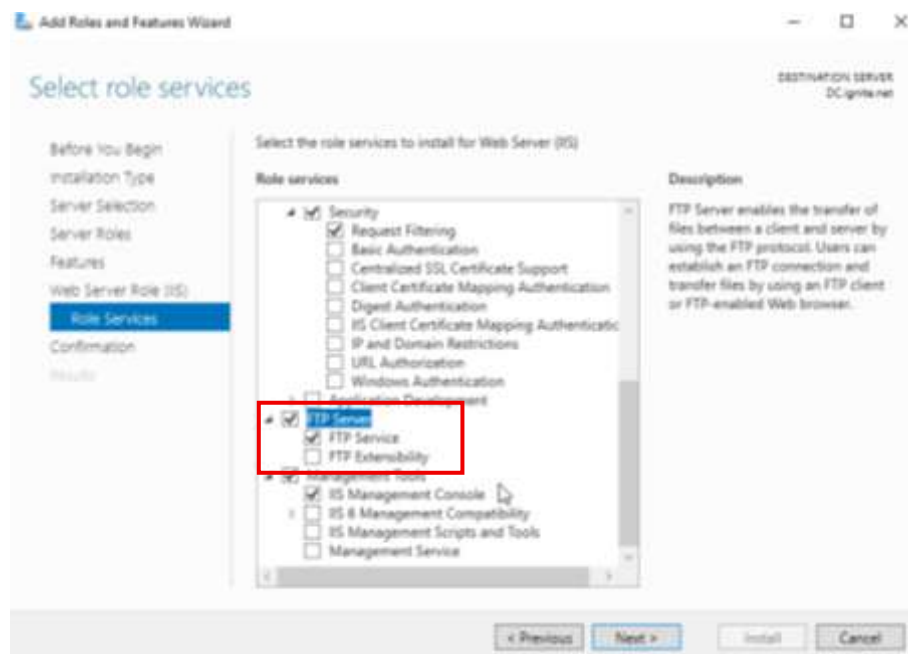
Configuring the Vulnerable FTP Server on DC

3.1.1 Install FTP Server Role:

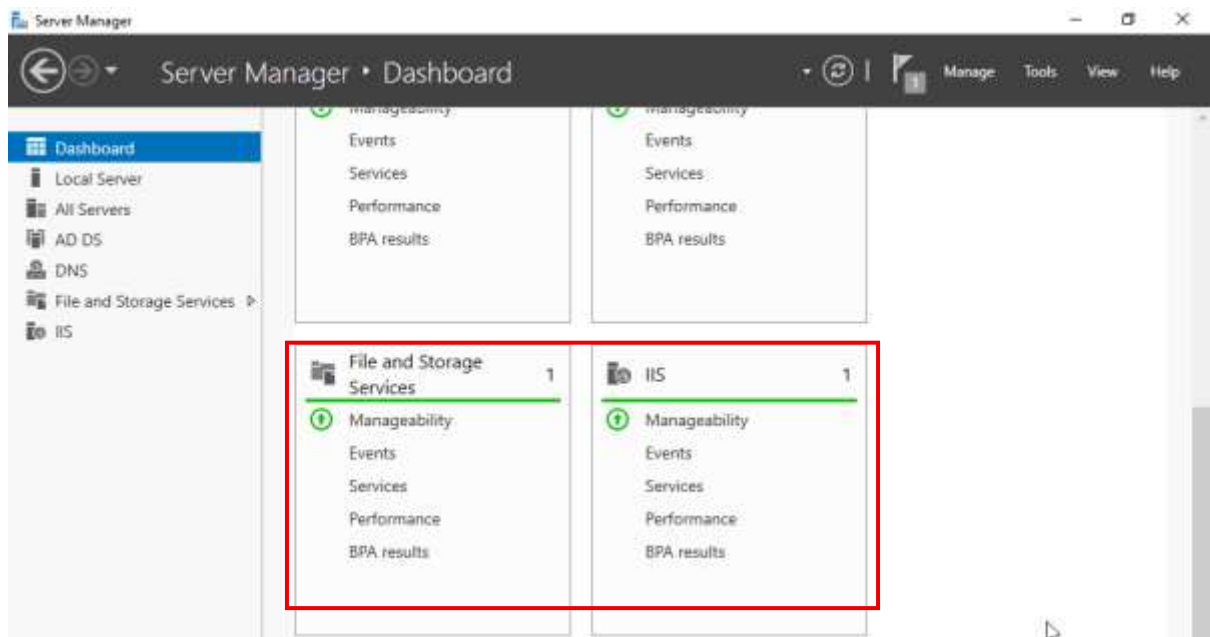
In Server Manager, add the Web Server (IIS) role.



Ensure FTP Service is selected under the Web Server role.



IIS has been added to the dashboard:

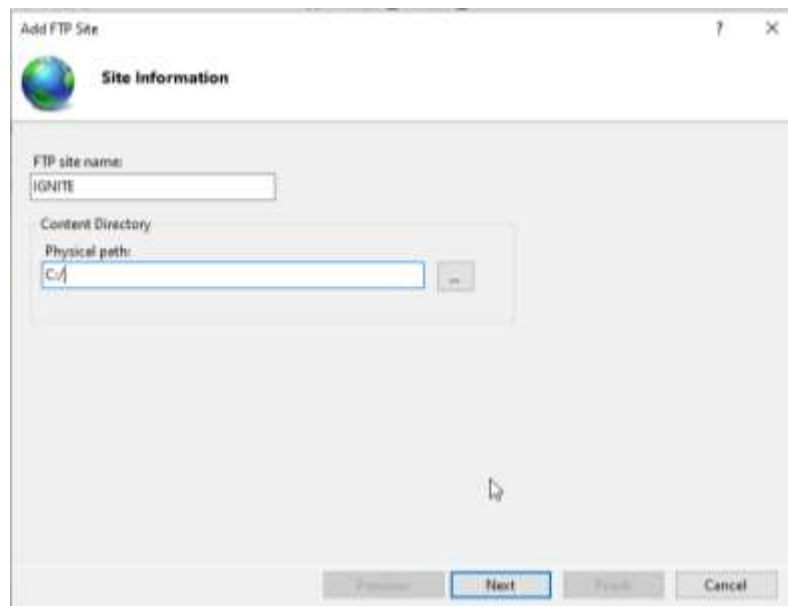


3.1.2 Create FTP Site (3.1.a):

Open IIS Manager → Right-click Sites → Add FTP Site

Name: Ignite

Physical Path: C:\



3.1.3 Bind & Auth:

Binding: Set IP Address to 192.168.20.1 and Port to 21.

SSL: Select No SSL. (Creates the vulnerability)

Add FTP Site ? X

Binding and SSL Settings

Binding

IP Address: 192.168.20.1 Port: 21

☐ Enable Virtual Host Names

Virtual Host (example: ftp.confhost.com):

☒ Start FTP site automatically

SSL

☒ No SSL

☐ Allow SSL

☐ Require SSL

SSL Certificate: Not Selected [Select...] [View...]

[Previous] [Next] [Finish] [Cancel]

Authentication: Select Basic only.

Authorization: Select All users with Read and Write permissions.

Add FTP Site ? X

Authentication and Authorization Information

Authentication

☐ Anonymous

☒ Basic

Authorization

Allow access to:

Specified users

Administrator

Permissions

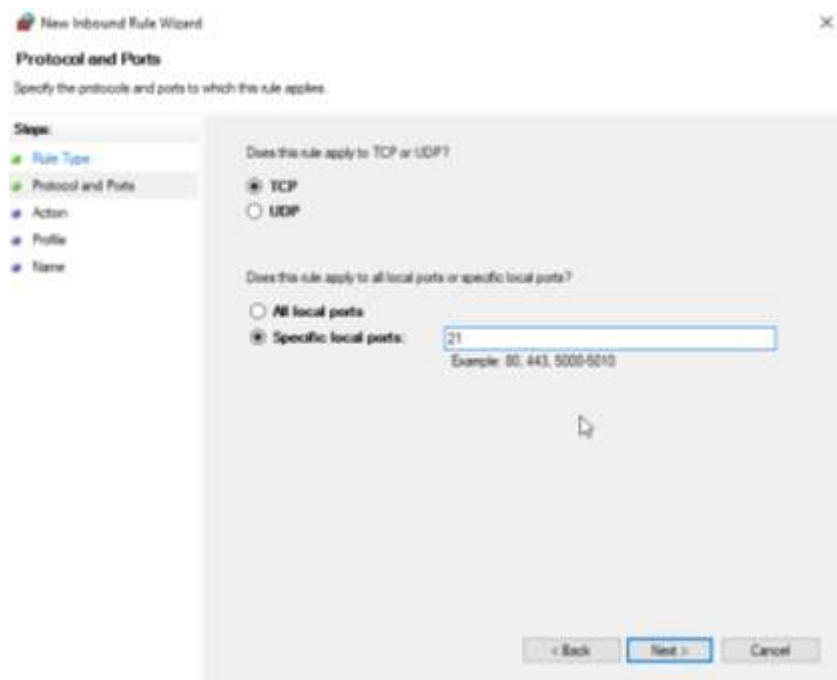
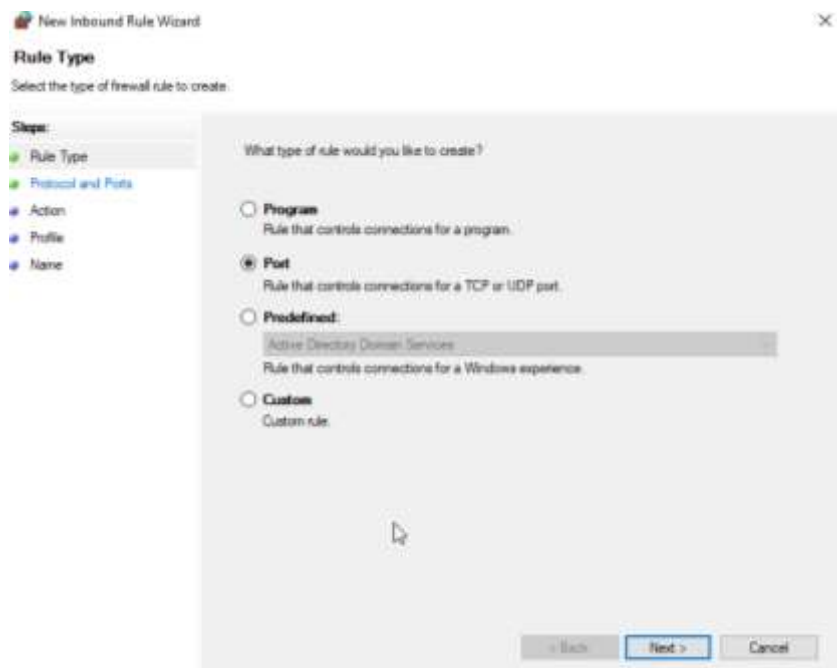
☒ Read

☒ Write

[Previous] [Next] [Finish] [Cancel]

3.2 Configure Firewall Policy:

Create an Inbound Rule in Windows Firewall to allow TCP Port 21 traffic.



Inbound Rules					
Name	Group	Profile	Enabled	Action	
✓ <u>Allow inbound tcp port 21 - FTP Control</u>		Domai...	Yes	Allow	
✓ Active Directory Domain Controller - Ec...	Active Directory Domain Ser...	All	Yes	Allow	
✓ Active Directory Domain Controller - Ec...	Active Directory Domain Ser...	All	Yes	Allow	

Reconnaissance and Brute Force Attack from Kali

3.3.1 Start Metasploit: Launch the console: msfconsole.


```
GNU nano 8.6                                     passwords.txt *
password, Password123, P@ssword!
```

3.3.4 Setup Brute Force:

```
"use auxiliary/scanner/ftp/ftp_login
set RHOSTS 192.168.20.1
set USER_FILE /path/to/users.txt
set PASS_FILE /path/to/passwords.txt"
```

3.3.5 Execute Brute Force:

"run"

```
msf auxiliary(scanner/ftp/ftp_login) > use auxiliary/scanner/ftp/ftp_login
msf auxiliary(scanner/ftp/ftp_login) > set RHOSTS 192.168.20.1
RHOSTS => 192.168.20.1
msf auxiliary(scanner/ftp/ftp_login) > set USER_FILE /home/princebardock/users.txt
USER_FILE => /home/princebardock/users.txt
msf auxiliary(scanner/ftp/ftp_login) > set PASS_FILE /home/princebardock/passwords.txt
PASS_FILE => /home/princebardock/passwords.txt
msf auxiliary(scanner/ftp/ftp_login) > run
[*] 192.168.20.1:21 - 192.168.20.1:21 - Starting FTP login sweep
[!] 192.168.20.1:21 - No active DB -- Credential data will not be saved!
[!] 192.168.20.1:21 - 192.168.20.1:21 - LOGIN FAILED: Administrator, Guest, FTPuser:password, Password123, P@ssword! (Incorrect: )
[*] 192.168.20.1:21 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

3.3.c Analyse Results:

The successful combination is **Administrator** and **Password123**.

3.4 Explain how you would defend against this type of attack

I would recommend adding a robust Account Lockout Policy through Group Policy on the Domain Controller. For example, lock the account after 5 failed login attempts for 30 minutes. Implement strict password policies aimed at increasing complexity and length that slow down dictionary/brute-force attacks.

Disable the unencrypted FTP (port 21). Move to FTPS, which allows encryption of credentials and data transfer over TLS/SSL.

Place the FTP server on a dedicated, secured DMZ network segment. Implement strict Inbound and Outbound Filtering on the firewall, allowing the FTP server to communicate only with needed hosts and segregating it from the core AD infrastructure.

Integrate Windows Server Security Event Logs, particularly for failed login events into a SIEM. Configure SIEM to generate an alert or a firewall block rule for a high volume of failed logins from the same source IP within a short time span, which would be a signature for brute-force attacks.

Question 4 - Vulnerability Scanning

4.1 Install and Activate Nessus Essential

Download the Nessus Essentials installer for Windows Server.

- Installer start screen:



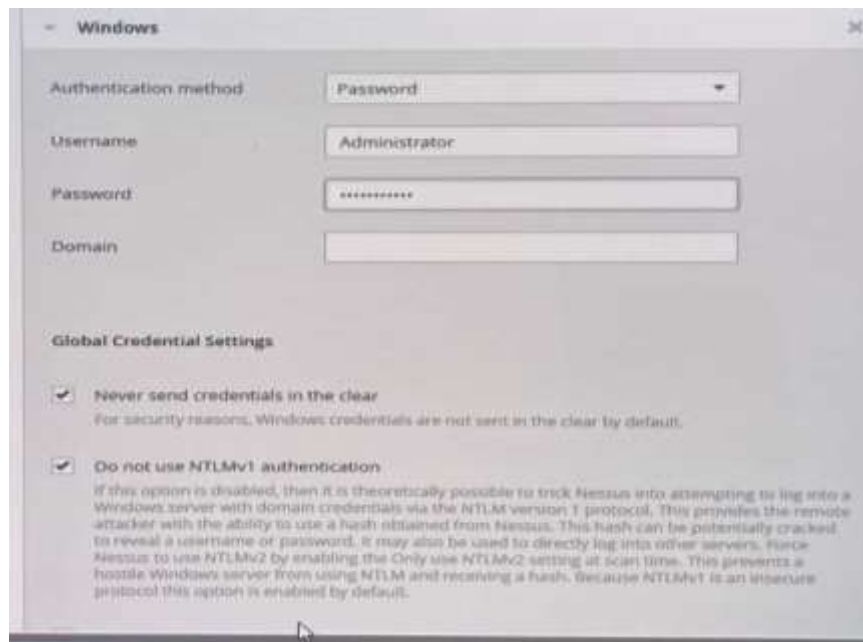
- Use your email to request the one-time activation key from Tenable:



- Use the key during the setup/web configuration phase:

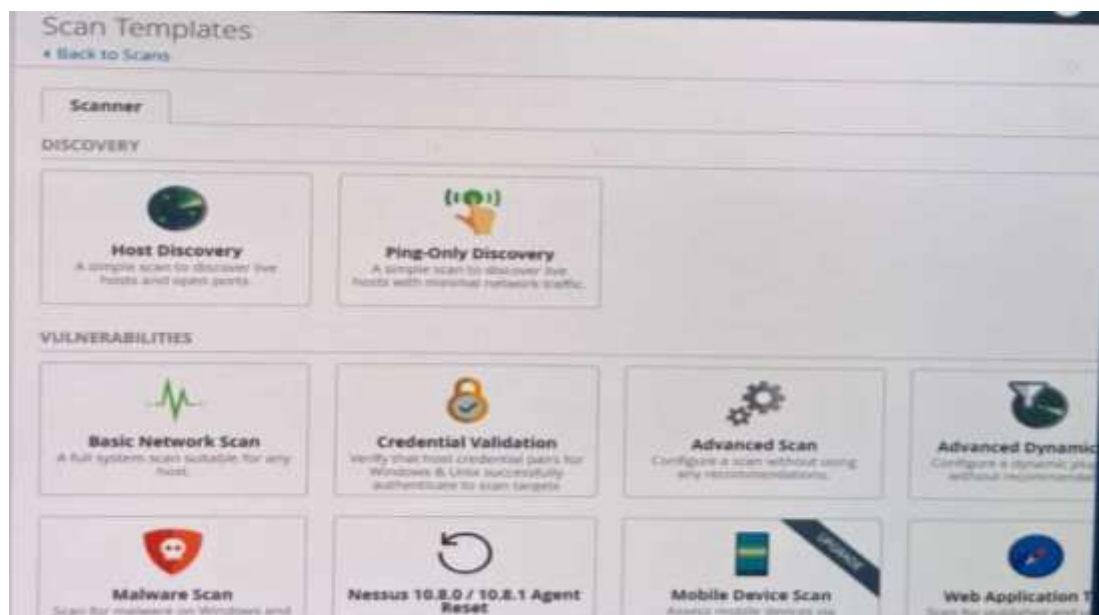


- Create the Nessus administrator account:



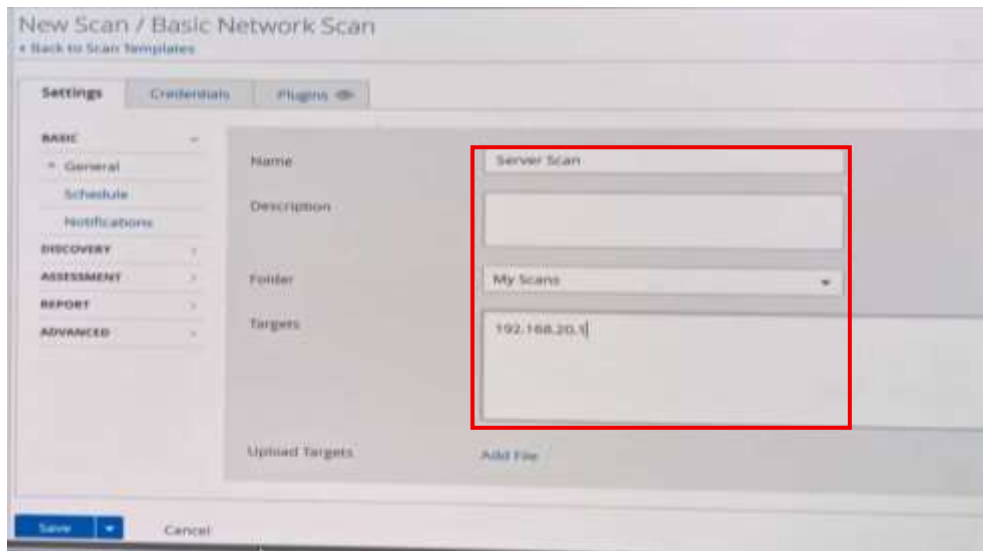
The screenshot shows a Windows authentication dialog box titled "Windows". It contains the following fields and settings:

- Authentication method:** A dropdown menu set to "Password".
- Username:** A text box containing "Administrator".
- Password:** A text box with masked characters (dots).
- Domain:** An empty text box.
- Global Credential Settings:**
 - ☒ **Never send credentials in the clear**
For security reasons, Windows credentials are not sent in the clear by default.
 - ☒ **Do not use NTLMv1 authentication**
If this option is disabled, then it is theoretically possible to trick Nessus into attempting to log into a Windows server with domain credentials via the NTLM version 1 protocol. This provides the remote attacker with the ability to use a hash obtained from Nessus. This hash can be potentially cracked to reveal a username or password. It may also be used to directly log into other servers. Force Nessus to use NTLMv2 by enabling the Only use NTLMv2 setting at scan time. This prevents a hostile Windows server from using NTLM and receiving a hash. Because NTLMv1 is an insecure protocol this option is enabled by default.



4.2 & 4.3 Create and Run the Scan

Select the New Scan option in the Nessus interface:

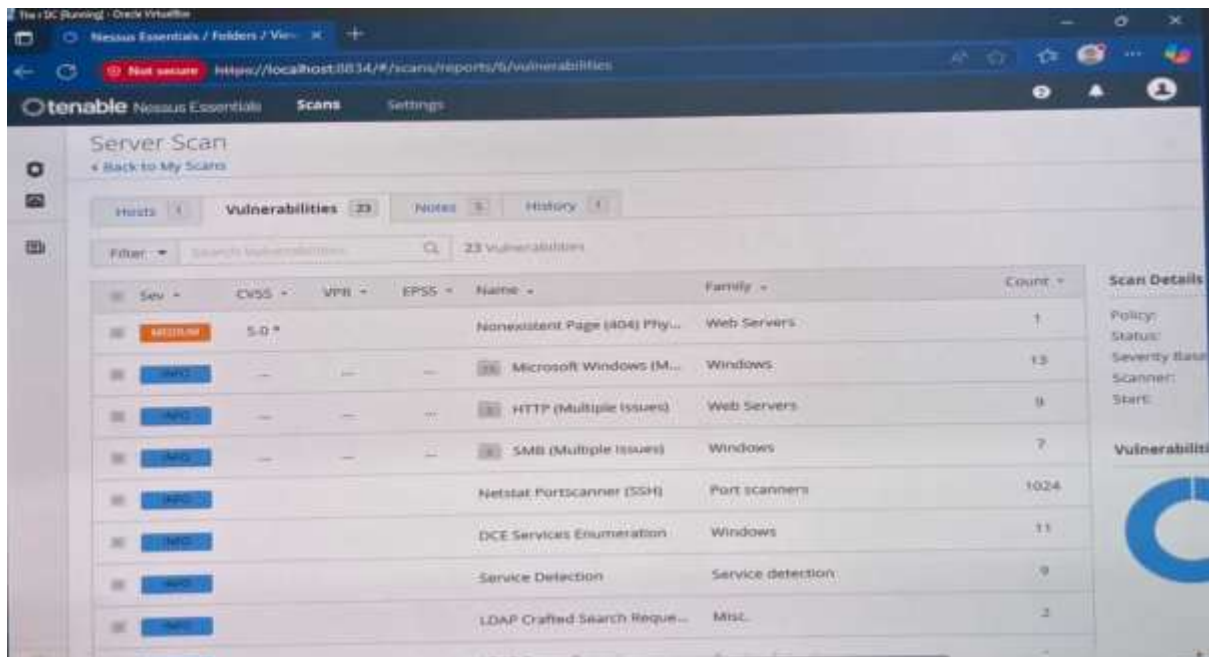


- Choose the Basic Network Scan template as specified in the question.
- Set the name exactly as "Server Scan".
- Input the specific IP address of your Windows Server 2022 VM.
- Save and launch the scan.

4.4 Analyse the Scan Results and Remediation

4.4a Analysis of Vulnerabilities

Filter the results by severity: Critical, High, Medium:



Select Top Findings: Identify and list 3-4 specific vulnerabilities (e.g., missing patches, insecure configurations, unnecessary open services).

- Microsoft Windows
- From the Web servers Family
- Medium Severity
- CVSS 5.0

- Allows unauthenticated remote code execution

4.4b Remediation Plan

Any solutions to issues should be dealt with using the severity and CVSS urgency order. Critical scores should be prioritised.

Ideally, we should apply all pending OS security patches and initiate Windows updates.

Next, unnecessary ports should be closed by disabling unused services.

- Ensuring local security policies are reviewed and strengthened prevents any massive threats from slipping through and deters weaker attacks. Local security policies include complex password requirements or account lockout thresholds

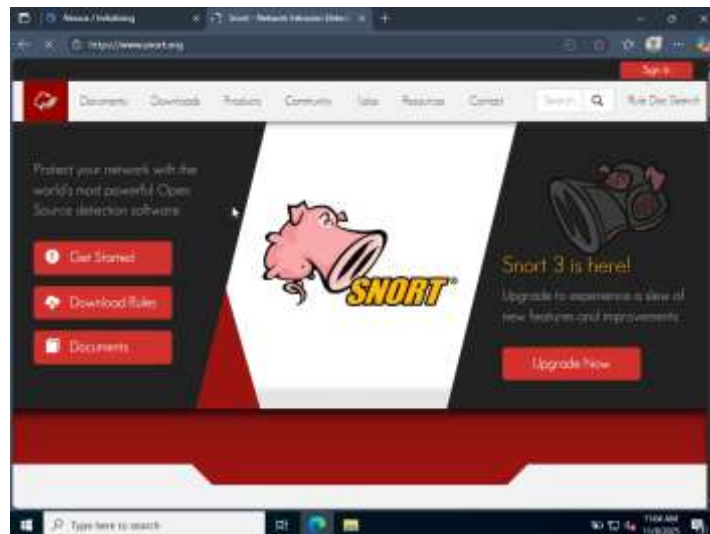
4.5 Explain how a vulnerability scan is useful to a penetration test

This is the main tool for discovery, mapping of the network to quickly identify all services and potential vulnerabilities. It frees the highly qualified resource-the penetration tester to not have to look for the "low-hanging fruit" and commonly known vulnerabilities. The output of the scan-either CVE identifiers or versions of software provides the exact information the pen tester needs for choosing the right exploit and will result in a very focused and probably effective attack phase. It provides instant, verifiable data on the organization's current risk posture well before any complex exploit attempts are made.

Question 5 - Intrusion Detection System

5.1 Install Snort

Download the Snort Installer for Windows from the official Snort website.



Download and install Npcap

5.2 Copy Resource Folders

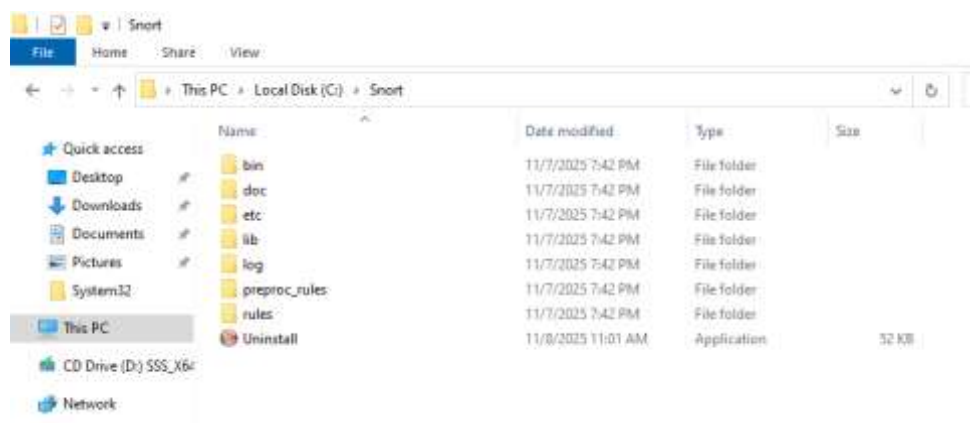
Identify the snort_rules resource folder provided for the assignment.

Navigate to your default Snort installation folder (C:\Snort).

Execute Copy and Replace:

Copy the entire contents (the folders like rules, preproc_rules, etc, etc.) from the snort_rules resource folder.

Paste them directly into the root of the Snort installation folder (C:\Snort).



5.3 Configure Network Settings in snort.conf

Navigate to the Snort configuration directory (C:\Snort\etc) and open the snort.conf file using a professional text editor (like Notepad)

Identify Network Variables:


```

snort.conf - Notepad
File Edit Format View Help
-----
# VRT Rule Packages Snort.conf
#
# For more information visit us at:
#   http://www.snort.org           Snort Website
#   http://vrt-blog.snort.org/     Sourcefire VRT Blog
#
# Mailing list Contact:   snort-users@lists.snort.org
# False Positive reports: fp@sourcefire.com
# Snort bugs:            bugs@snort.org
#
# Compatible with Snort Versions:
# VERSIONS : 2.9.20
#
# Snort build options:
# OPTIONS : --enable-gre --enable-apis --enable-targetbased --enable-gpm --enable-perfprofiling --enable-zlib --enable-ac
#
# Additional Information:
# This configuration file enables active response, to run snort in
# test mode -T you are required to supply an interface -i <interface>
# or test mode will fail to fully validate the configuration and
# exit with a FATAL error
#-----
#####
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder
# 3) Configure the base detection engine
# 4) Configure dynamic loaded libraries
# 5) Configure preprocessors
# 6) Configure output plugins
# 7) Customize your rule set
# 8) Customize preprocessor and decoder rule set
#-----

```

Update the ipvar HOME_NET line.

Replace the default variable with your Server's specific IP address.

```

#####
# Step #1: Set the network variables. For more information, see README.variables
#####

# Setup the network addresses you are protecting
ipvar HOME_NET 192.168.20.1/32

```

Configure EXTERNAL_NET: Update the ipvar EXTERNAL_NET line.

Set this variable to represent everything that is not your home network.

```

# Set up the external network addresses. Leave as "any" in most situations:
ipvar EXTERNAL_NET !$HOME_NET

```

Save and close the snort.conf file.

5.4 Configure local.rules

Download Snort rules



5.4.a. FTP Connectivity Alert

This rule uses the default FTP port (21) as a clear indicator of that protocol's use towards the protected asset.

Rule Type: Standard Snort Alert Rule.

```
"alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected";
sid:1000001; rev:1;)"

# site specific rules
include $RULE_PATH/local.rules

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1;)
```

Breakdown:

alert tcp: Action (alert) on Protocol (TCP).

any any: Source IP (any) and Source Port (any).

->: Direction operator (inbound to the protected network).

\$HOME_NET 21: Destination IP (your server) and Destination Port (FTP).

(msg:..."; sid:1000001; rev:1;): The required alert message and mandatory unique Snort Rule ID (SID) and revision number (Rev).

5.4.b. FIN Scan Alert

This rule detects reconnaissance activity, specifically a stealthy FIN scan, by checking the TCP header flag.

Rule Type: Alert Rule with the crucial flags option.

```
"alert tcp any any -> $HOME_NET any (msg:"Server Services Scanned!!"; flags:F;
sid:1000002; rev:1;)"

# site specific rules
include $RULE_PATH/local.rules

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET any (msg:"Server Services Scanned!!"; flags:F; sid 1000002; rev:1;)|
```

Breakdown:

flags:F: This is the key component. It instructs Snort to only fire this alert if the TCP header has only the FIN flag set.

5.4.c. Limit Output (Rate Limiting)

This is crucial Security Intelligence Tuning to prevent an attacker from flooding your log with alerts (TDoS). You are setting a threshold of 5 alerts per 5 seconds per source.

```
"detection_filter: track by_src, count 5, seconds 5;"

# site specific rules
include $RULE_PATH/local.rules

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET any (msg:"Server Services Scanned!!"; flags:F; sid 1000002; rev:1;)

detection_filter: track by_src, count 5, seconds 5;
```

Final Rule 5.4.a:

"alert tcp any any -> \$HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1; detection_filter: track by_src, count 5, seconds 5;)"

```
# site specific rules
include $RULE_PATH/local.rules

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET any (msg:"Server Services Scanned!!"; flags:F; sid 1000002; rev:1;)

detection_filter: track by_src, count 5, seconds 5;

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1; detection_filter: track by_src, count 5, seconds 5;)
```

Final Rule 5.4.b:

"alert tcp any any -> \$HOME_NET any (msg:"Server Services Scanned!!"; flags:F; sid:1000002; rev:1; detection_filter: track by_src, count 5, seconds 5;)"

```
# site specific rules
include $RULE_PATH/local.rules

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET any (msg:"Server Services Scanned!!"; flags:F; sid 1000002; rev:1;)

detection_filter: track by_src, count 5, seconds 5;

alert tcp any any -> $HOME_NET 21 (msg:"FTP Connectivity Detected"; sid:1000001; rev:1; detection_filter: track by_src, count 5, seconds 5;);

alert tcp any any -> $HOME_NET any (msg:"Server Services Scanned!!"; flags:F; sid:1000002; rev:1; detection_filter: track by_src, count 5, seconds 5;);
```

Save and close the local.rules file.

5.5 Run Snort to Capture Data

Launch the Command Prompt as an Administrator.

The Snort engine needs to know which network card to listen on.

Run the interface listing command:

"snort -W"

Run the program with the necessary configuration and output flags.

"snort -A console -c C:\Snort\etc\snort.conf -i <Interface_Number> -l C:\Snort\log -K ascii"



Breakdown:

-A console: Instructs Snort to send alerts to the console (standard output), meeting the assignment requirement.

-c <path>: Specifies the path to your configured snort.conf file.

-i <Interface_Number>: Specifies the interface number you identified with snort -W.

-I <path>: Sets the logging directory for full packet dumps (best practice, even if you are using console output).

-K **ascii**: Ensures logging is done in a readable text format.

Initiating an FTP Connection

Step 1: Open Kali Terminal.

Step 2: Use the standard FTP client to initiate a connection to the target server IP.

“sudo nmap ftp 192.168.20.1”

```
(princehardock@kali)-[~]
└─$ sudo nmap ftp 192.168.20.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-08 20:18 PST
Failed to resolve "ftp".
Nmap scan report for 192.168.20.1
Host is up (0.0062s latency).
Not shown: 984 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
53/tcp    open  domain
80/tcp    open  http
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
3389/tcp  open  ms-wbt-server
5357/tcp  open  wsdapi
5985/tcp  open  wsman
MAC Address: 08:00:27:57:99:62 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.79 seconds
```

Step 3: Monitor Snort. Immediately look at the Windows Server console for the custom alert message.

Snort: *[] [1:1000001:1] FTP Connectivity Detected [**]***

```
Select C:\Snort\bin\snort.exe
=====
11/08-13:49:04.456277 192.168.20.1:111 -> 192.168.20.2:53415
TCP TTL:128 TOS:0x0 ID:14268 Iplen:20 DgmLen:40 DF
***A**R** Seq: 0x0 Ack: 0xFF4CEF1F Win: 0x0 TcpLen: 20
=====
11/08-13:49:04.456510 192.168.20.1:587 -> 192.168.20.2:53415
TCP TTL:128 TOS:0x0 ID:14269 Iplen:20 DgmLen:40 DF
***A**R** Seq: 0x0 Ack: 0xFF4CEF1F Win: 0x0 TcpLen: 20
=====
11/08-13:49:04.456702 192.168.20.1:1720 -> 192.168.20.2:53415
TCP TTL:128 TOS:0x0 ID:14270 Iplen:20 DgmLen:40 DF
***A**R** Seq: 0x0 Ack: 0xFF4CEF1F Win: 0x0 TcpLen: 20
=====
11/08-13:49:04.456926 192.168.20.1:25 -> 192.168.20.2:53415
TCP TTL:128 TOS:0x0 ID:14271 Iplen:20 DgmLen:40 DF
***A**R** Seq: 0x0 Ack: 0xFF4CEF1F Win: 0x0 TcpLen: 20
=====
11/08-13:49:04.457115 192.168.20.1:3306 -> 192.168.20.2:53415
TCP TTL:128 TOS:0x0 ID:14272 Iplen:20 DgmLen:40 DF
***A**R** Seq: 0x0 Ack: 0xFF4CEF1F Win: 0x0 TcpLen: 20
=====
11/08-13:49:04.457362 192.168.20.1:443 -> 192.168.20.2:53415
TCP TTL:128 TOS:0x0 ID:14273 Iplen:20 DgmLen:40 DF
***A**R**
```

Step 4: Type quit to exit the FTP shell on Kali once the alert is confirmed.

Initiating a FIN Scan

Step 1: Open Kali Terminal.

Step 2: Execute the Nmap FIN Scan. Use the dedicated -sF flag for this type of stealth scan.

"nmap -sF 192.168.20.1"

```
(princebardock@kali)~$ sudo nmap -sF 192.168.20.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-08 20:19 PST
Nmap scan report for 192.168.20.1
Host is up (0.0055s latency).
All 1000 scanned ports on 192.168.20.1 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:57:99:62 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds
```

Step 3: While the Nmap scan is running, observe the Windows Server console.

Snort: Multiple of [**] [1:1000002:1] Server Services Scanned!! [**]

Step 4: After the initial 5 alerts, the output should momentarily stop for 5 seconds (from the same source IP), even though the scan is still running. This confirms your tuning rule `detection_filter: track by_src, count 5, seconds 5;` is working effectively.

Reference List:

<https://www.nist.gov/publications/guide-intrusion-detection-and-prevention-systems-idps>

<https://csrc.nist.gov/library/NIST%20SB%20200702%20Intrusion%20Detection%20And%20Prevention%20Systems.pdf>

<https://www.snort.org/>

<https://docs.suricata.io/en/suricata-8.0.2/>

<https://www.sans.org/white-papers>

<https://docs.tenable.com/nessus/Content/CredentialedChecksOnWindows.htm>

<https://docs.tenable.com/vulnerabilitymanagement/bestpractices/security/Content/VulnerabilityPriorityRating.htm>

<https://docs.tenable.com/cyberexposurestudies/ransomware/Content/PrioritizeUsingPrediction.htm>

<https://docs.tenable.com/pdfs/VPR-enhancements-FAQ/Content/VPR/Enhancements.htm>

<https://www.youtube.com/watch?v=9kmotUhDhZg>

https://www.cisecurity.org/benchmark/microsoft_windows_server

<https://www.cisecurity.org/cis-benchmarks>

https://www.tenable.com/audits/items/CIS_MS_Windows_10_Enterprise_Level_1_v1.5.0.audit:698d681eea19e6c20075617030f95ce6

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<https://attack.mitre.org/matrices/enterprise/>

https://owasp.org/wwwprojectwebsecuritytestingguide/stable/4Web_Application_Security_Testing/01-Information_Gathering/

<https://techdocs.broadcom.com/us/en/vmware-cis/desktop-hypervisors/workstation-pro/25H2/using-vmware-workstation-pro/introduction-and-system-requirements/virtual-machine-requirements/virtual-networking-requirements-for-virtual-machines.html>

<https://www.virtualbox.org/manual/ch06.html>
